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H2**Started on** Thursday, 24 September 2026, 9:13 AM**State** Finished**Completed on** Thursday, 24 September 2026, 9:18 AM**Time taken** 4 mins 57 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int A[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &A[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (j < n) {
18         int diff = A[j] - A[i];
19
20         if (diff == k && i != j) {
21             printf("1\n");
22             return 0;
23         }
24         else if (diff < k) {
25             j++;
26         }
27         else {
28             i++;
29         }
30
31         if (i == j) {
32             j++;
33         }
34     }
35
36     printf("0\n");
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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